

In theory, an audio wave can range from 20Hz to 20,000Hz. For AM (amplitude modulation) based radio communication, a carrier wave is needed to transmit the audio wave. At the radio transmitter side, we take the audio wave, which has a changing frequency and amplitude, and modify the amplitude of the carrier wave according to the audio wave, while keeping the frequency of the carrier wave unchanged. The resulting output, or the modified carrier wave, is called a modulated wave. Note that only the amplitude of the carrier is modified, hence the name amplitude modulation (AM). Essentially, the modulated wave is then sent to the antenna for AM broadcasting.

Why do we need a carrier wave? It is a good question. Even though it is not very practical, let us just imagine that we were trying to send the audio signal directly for AM broadcasting. How do we then create a circuit that will pick up the radio signal from a particular radio station if there are dozens of them in an area? If all the AM radio stations in a city broadcast the audio directly, we will not be able to distinguish them from each other, because all would be transmitting anywhere and everywhere between 20Hz and 20,000Hz! That would be like everyone in a room full of a hundred people shouting at each other in the loudest way and no one being able to hear clearly what any other person was saying.

To prevent this issue, it was thought in the early days that every AM radio station should be given a fixed frequency to transmit so that any receiver tuning to that frequency would be able to hear only that station. The frequency on which a particular radio station is transmitting is called the carrier frequency of that station.

A radio station just sending a

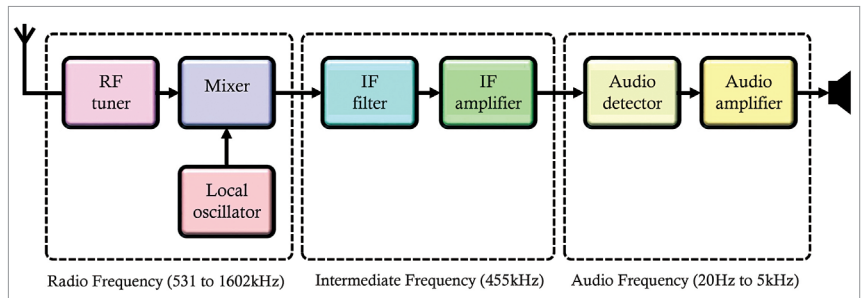


Fig. 1: A simplified diagram of an AM radio receiver

How IF works in radio receivers

In a typical radio receiver, the incoming radio frequency (RF) signal is mixed with a signal from a local oscillator. This mixing process generates two new frequencies: the sum and difference of the original frequencies. The difference frequency is usually the intermediate frequency (IF). By converting the original RF signal to this IF, it becomes easier to process - filter, amplify, and demodulate.

carrier wave of a constant amplitude and frequency is of no use because the receiver will not get any audio out of it. Remember the modulation? Essentially, modifying the amplitude of such carrier wave with an audio signal and then sending that from the radio station will make it useful (see Fig. 2). Yes, it is still at the frequency fixed for the radio station as we modified only the amplitude and not the frequency. Thus, we could send the audio signal by means of an amplitude change of the carrier wave to the receiving radios tuned to that carrier frequency.

For Europe, Africa, and Asia, the medium wave AM band consists of carrier frequencies from 531kHz to 1602kHz, with each station separated by 9kHz. (For a deeper look at

the evolution of radio receivers, see the article titled 'The Transformation of Radio Receivers' at <https://www.electronicsforu.com/market-verticals/radio-receivers-transformation> link to understand how the radio receivers evolved.)

Challenges in AM reception

As we now know, each broadcasting station transmits its amplitude modulated carrier wave at its own specified frequency, as fixed by the concerned government authorities and regulations.

This is good. However, we still have a problem.

We now need to have a radio receiver designed with an amplifier for amplifying the received carrier wave. That is also not a problem.

The real problem is designing an amplifier that will amplify such high frequency, but the response of the amplifier should also be good for an entire range of possible carrier frequencies covering all the radio stations between 531kHz and 1602kHz!

Having an amplifier

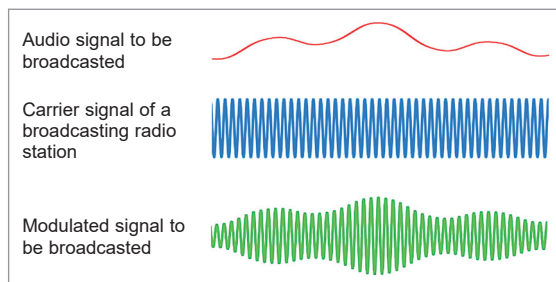


Fig. 2: At transmitter side, an audio signal is used to modify the amplitude (amplitude modulation) of another signal called carrier signal to generate a modulated signal to be broadcasted